PAPIT: Prompt-Aware Pruning for Image Tokens in Efficient Multimodal Inference

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Abstract

Multimodal Large Language Models (MLLMs) such as LLaVA encode images as hundreds of visual patch tokens, most of which are irrelevant to a given user query. A natural hypothesis is that CLIP-derived cross-modal saliency, which is already query-conditioned and requires no training, can identify the relevant tokens. We test this hypothesis with ***PAPIT*** *(**P**rompt-**A**ware **P**runing for **I**mage **T**okens), a training-free pipeline that scores visual patches against the user's prompt using GradCAM saliency on LLaVA's own vision tower and retains only the top-k most relevant tokens before LLaVA inference. Efficiency experiments confirm a 41–90% reduction in attention FLOPs with negligible scoring overhead. However, on three VQA benchmarks (700 samples each), PAPIT only matches a query-agnostic random baseline: GQA 69.7% vs. 70.1%, VQA v2 74.8% vs. 75.8%, TextVQA 42.9% vs. 42.0% at k=75%. To explain this, we directly measure the alignment between PAPIT's GradCAM scores and LLaVA's internal attention over the 576 image tokens on 100 samples per benchmark, and find on all three datasets a per-image Spearman* $\rho \approx 0$ *with top-*10% *Jaccard* below *the uniform-random baseline. Our main contribution is therefore a quantitative* negative result*: CLIP-aligned saliency carries essentially no information about which patches the LLM relies on, ruling out an entire family of training-free prompt-aware pruning approaches. As a positive control, we test an LLM-internal-attention oracle on the same pipeline and find that mid-layer attention (*$L \approx 16$ *in LLaVA-1.5-7B) does beat random (+0.08 at* $k = 25\%$*, matching unpruned at* $k = 50\%$ *on GQA-100), showing that prompt-aware pruning is achievable provided the signal is read from inside the LLM rather than from the CLIP-aligned vision side.*

1. Introduction

LLaVA-1.5 [1] processes a 336×336 image as a $24 \times 24 = 576$ patch token grid via a CLIP ViT-L/14 vision encoder [2, 3]. Each token participates in every self-attention layer [4] of the language model, making inference cost quadratic in the number of visual tokens. For a typical VQA query, fewer than half of these tokens are semantically relevant.

A natural intuition is that the user's prompt should be enough to identify which tokens to keep, *without* any training, since CLIP and its derivatives already provide a query-conditioned cross-modal embedding space. We ask: *can a training-free saliency signal, read from LLaVA's own CLIP-initialized vision tower, predict which patches the LLaVA LLM actually relies on, well enough to beat a query-agnostic random baseline?*

We instantiate this as **PAPIT**, a training-free pipeline that scores patches by GradCAM saliency on LLaVA's vision tower against the prompt's CLIP text embedding, retains the top-k patches plus a global anchor, and feeds the pruned set through LLaVA's MLP projector and LLM. Across three VQA benchmarks, PAPIT *ties* random pruning rather than beating it (Sec. 6), even though Figure 2 shows interpretably localised selection on individual examples. An alignment diagnostic (Sec. 7) on 100 samples from each of the three benchmarks explains why: per-image Spearman $\rho \approx 0$ between PAPIT's GradCAM scores and LLaVA's LLM attention over the 576 image tokens, and the top-10% Jaccard falls slightly *below* the uniform-random reference on every dataset.

Contributions. (i) A clean training-free token-pruning pipeline for LLaVA-1.5 with 41–90% attention-FLOPs reduction and ≤2.3-point accuracy loss vs. unpruned on GQA and VQA v2 at k=75%. (ii) A quantitative *negative result*: CLIP-derived patch saliency is statistically uncorrelated with LLaVA's LLM attention, which rules out an entire family of training-free prompt-

aware methods and motivates LLM-internal scoring (FastV-style attention, or trained predictors over LLaVA hidden states) for any future prompt-conditioned approach.

2. Related Work

Token pruning for Vision Transformers has been studied for image classification [5, 6], where a learned predictor module decides which tokens to drop. In the multimodal setting, FastV [7] prunes visual tokens based on attention scores inside the LLM, requiring a forward pass before pruning. A complementary line of work shows that dense patch-level signals can be extracted directly from CLIP without supervision, e.g. MaskCLIP [8], motivating the use of CLIP-derived saliency for query-conditioned scoring. PAPIT differs in that pruning happens *before* the LLM sees the image, using a lightweight cross-modal scoring step (≈85 ms, dominated by the CLIP text-encoder forward pass) that is entirely query-conditioned and requires no training.

3. Methodology

3.1. Pipeline Overview

Figure 1 shows the three-stage pipeline. PAPIT reuses LLaVA-1.5's frozen vision tower, which is itself a CLIP ViT-L/14, so no second image encoder is loaded; only the matching CLIP text encoder is added on top to obtain a text embedding for cross-modal scoring. The image is encoded by the (shared) ViT and the prompt by the CLIP text encoder; GradCAM scoring then selects the top-k most relevant patches; the pruned tokens are projected by LLaVA's MLP projector and fed to the LLM to generate an answer.

3.2. Cross-Modal Scoring: GradCAM Saliency

Because CLIP's contrastive objective aligns only the CLS token with text, naïve patch–text cosine similarity is unstable across retention ratios and collapses at k=50% (Table 3); GradCAM saliency on the vision tower provides a more reliable patch-level signal, as confirmed by the ablation in Section 6.

GradCAM scoring. Let $H^{(L-1)} = \{h_1, \ldots, h_n\}$ denote patch activations at the penultimate ViT encoder layer, and sim the cosine similarity between the projected CLS image embedding and the text embedding T. We attribute sim to each patch via GradCAM [9]:

$$s_i^{\mathrm{gc}} = \mathrm{ReLU}\left(\sum_d \frac{\partial\,\mathrm{sim}}{\partial\, h_{id}} \cdot h_{id}\right). \tag{1}$$

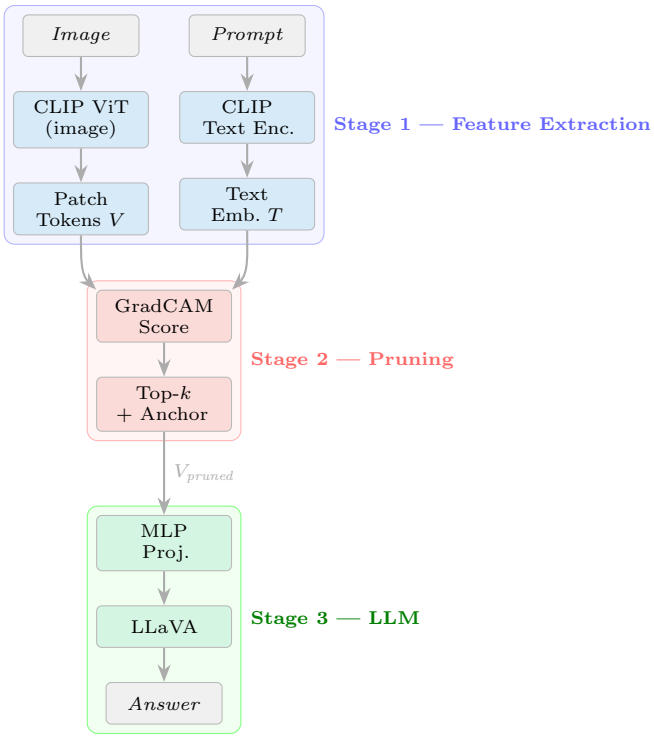

Figure 1. PAPIT pipeline. Pruning happens between the ViT and the LLaVA MLP projector, requiring no modification to the LLM weights.

Backpropagating through the final attention layer captures each patch's *causal* contribution to the global text-image alignment. The top-k patches are retained: $V_{\mathrm{pruned}} = \{h_i \mid s_i^{\mathrm{gc}} \in \mathrm{TopK}(\{s_j^{\mathrm{gc}}\}_{j=1}^{n}, k)\}$.

3.3. Spatial Anchoring

After top-k selection, the pruned tokens preserve their original 2-D grid indices as position IDs before being passed to the MLP projector. One *anchor token* (mean of all dropped tokens) is appended to provide global context, giving a final sequence of length $k + 1$.

3.4. Efficiency Reduction

Self-attention cost scales quadratically with sequence length. Let L_{text} be the number of text tokens (typically ≈50). Relative FLOPs compared to unpruned inference is:

$$\mathrm{FLOPs}_{\mathrm{rel}} = \frac{(k + L_{\mathrm{text}})^2}{(n + L_{\mathrm{text}})^2}. \tag{2}$$

For LLaVA-1.5 with $n = 576$: retaining $k = 25\%$ (144 tokens) yields $\mathrm{FLOPs}_{\mathrm{rel}} \approx 0.10$, a ∼10× reduction.

3.5. Implementation

PAPIT operates at the token level between the ViT encoder and the LLaVA MLP projector, requiring no modification to LLM weights. The image

branch reuses LLaVA-1.5's own vision tower (a frozen CLIP ViT-L/14); the text branch loads the matching `clip-vit-large-patch14` text encoder via HuggingFace Transformers, so the only extra parameters introduced by PAPIT are CLIP's text encoder weights.

4. Qualitative Results

Figure 2 illustrates prompt-aware patch selection on two GQA examples. In the first example, GradCAM highlights the player region in response to a jersey-colour query, and PAPIT's retained patches are visibly more concentrated on the subject than random selection. In the second example, the strong colour contrast between the orange umbrella and blue sky produces a clear saliency map: GradCAM assigns high scores to the umbrella boundary, and PAPIT's selection concentrates on the umbrella area while random selection scatters across the irrelevant sky, demonstrating a case where prompt-aware scoring provides an interpretable spatial advantage.

5. Quantitative Results

We conducted an efficiency benchmark using a BLIP [10] VQA proxy model (Table 1) and a full LLaVA-1.5 accuracy evaluation on GQA [11], VQA v2 [12], and TextVQA [13] (Table 2). The proxy model uses a $7 \times 7 = 49$ patch grid; full LLaVA experiments use $n = 576$.

Table 1. Efficiency benchmark results (BLIP proxy model, single A6000 GPU, averaged over 3 runs per setting). Prune latency includes CLIP scoring + top-k selection. E2E latency includes pruning and the BLIP forward pass.

k	Tokens kept	Keep ratio	$\text{FLOPs}_{\text{rel}}$ (Eq. 2)	Prune (ms)	E2E (ms)
100%	49	1.000	1.000	104	301
75%	37	0.755	0.772	81	276
50%	24	0.490	0.559	95	291
25%	12	0.245	0.392	82	247

Pruning latency is approximately constant at ≈85 ms regardless of k, dominated by the CLIP text encoder forward pass, confirming that the scoring overhead is negligible. E2E latency for all pruned variants ($k \leq 75\%$) remains below 300 ms, demonstrating practical wall-clock savings on the proxy model.

6. Experiments

Table 2 shows the accuracy evaluation. We compare three variants at each retention ratio using LLaVA-1.5-7B on 700 samples per dataset: **Unpruned** (all 576 patches), **Random** (uniformly sampled k patches), and **PAPIT** (GradCAM scoring, Section 3). Table 3 shows a scoring method ablation on GQA (300 samples) comparing three training-free strategies.

Table 2. Accuracy results on LLaVA-1.5-7B (700 samples each). VQA soft accuracy: min(count/3, 1) over 10 annotations; exact match for GQA (single answer). Rel. FLOPs from Eq. 2 (n=576, L_{text}≈50).

Dataset	Method	k=25%	k=50%	k=75%
Rel. FLOPs (Eq. 2)		*0.10*	*0.29*	*0.59*
GQA	Unpruned	72.0 (shared baseline)		
	Random	65.9	69.1	70.1
	PAPIT	65.1	68.7	69.7
VQA v2	Unpruned	76.5 (shared baseline)		
	Random	71.9	74.6	75.8
	PAPIT	70.9	74.5	74.8
TextVQA	Unpruned	49.0 (shared baseline)		
	Random	32.3	39.8	42.0
	PAPIT	31.6	38.4	**42.9**

Table 3. Scoring method ablation on GQA (LLaVA-1.5-7B, 300-sample subset; the main results in Table 2 use 700 samples, which explains the slight baseline shift). Unpruned: 71.3%. *Cosine+GradCAM hybrid* thresholds patch–text cosine similarity and falls back to GradCAM when the threshold yields too few patches; *Value features* scores patches by the norm of their value projections in the final ViT attention block.

Scoring method	k=25%	k=50%	k=75%
Cosine + GradCAM hybrid	65.3	53.7	58.3
Value features	58.3	64.3	68.3
GradCAM (Eq. 1)	**65.3**	**68.3**	**70.7**
Random baseline	64.3	68.7	69.3

Across all three datasets and retention ratios, PAPIT consistently achieves accuracy within ∼1% of random pruning, while reducing attention FLOPs by 41–90%. On GQA, PAPIT reaches 69.7% at k=75% compared to random 70.1% (−0.4 points) and unpruned 72.0% (−2.3 points total). On VQA v2, PAPIT reaches 74.8% vs random 75.8% at k=75%. On TextVQA, PAPIT slightly *surpasses* random at k=75% (42.9% vs 42.0%, +0.9 points). This is the one setting where prompt-aware scoring provides a measurable advantage: text regions are small and spatially concentrated, so GradCAM's saliency signal is more discriminative when a large fraction of patches is retained. Both methods remain well below the unpruned baseline (49.0%), reflecting the difficulty of recovering all text-region patches under pruning.

The ablation (Table 3) shows that cosine similarity with a fixed threshold collapses at k=50% (53.7%),

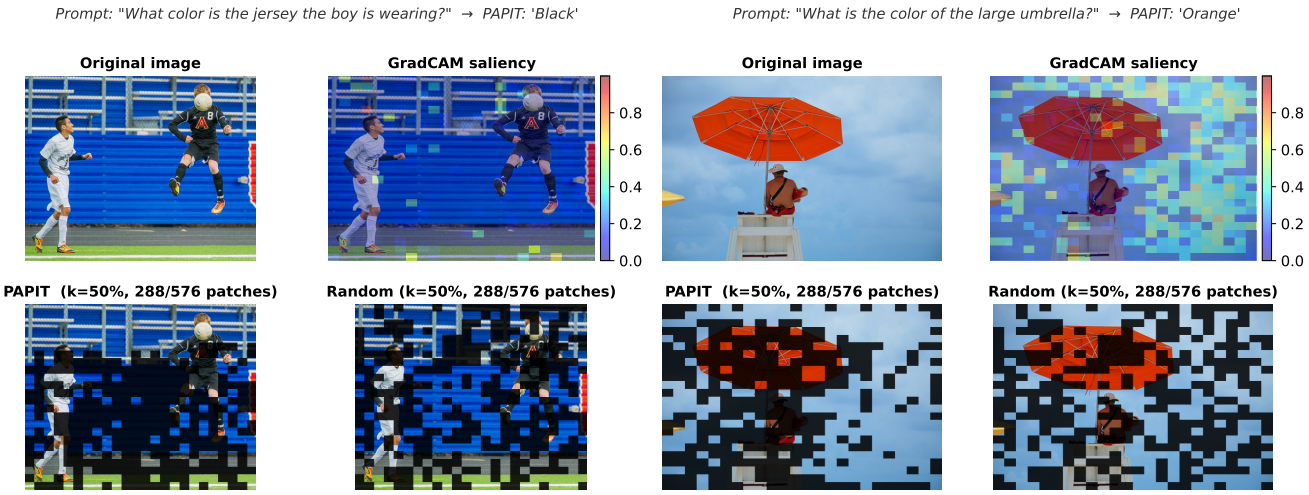

Figure 2. **Prompt-aware patch selection** on two GQA examples (k=50%). *Top:* original image and GradCAM saliency (warm = high score). *Bottom:* PAPIT-selected vs. randomly selected patches (black = dropped). PAPIT preserves semantically relevant regions; random selection scatters uniformly.

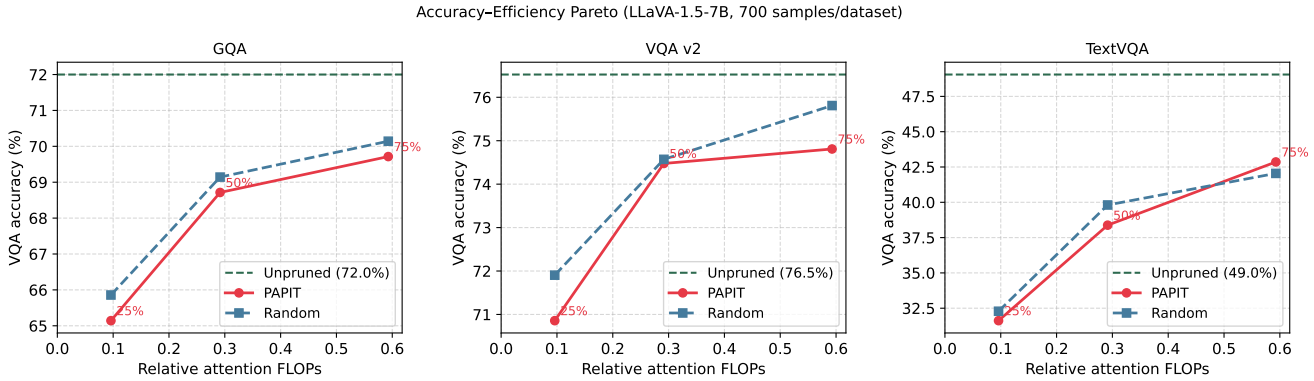

Figure 3. Accuracy–Efficiency Pareto across three benchmarks (LLaVA-1.5-7B, 700 samples each). PAPIT (red solid) closely tracks random pruning (blue dashed) across all retention ratios and datasets, with both methods recovering most unpruned accuracy (green dashed) at k=75% (∼41–90% FLOPs reduction).

confirming that raw feature-space alignment between LLaVA's ViT and CLIP's text encoder is unreliable. At k=25% the cosine threshold typically retains fewer than 25% of the patches, so the hybrid falls back entirely to GradCAM and matches the pure-GradCAM score (65.3%). Value-feature scoring also underperforms random across all retention ratios. GradCAM, which backpropagates through the CLS-to-text similarity, provides a more stable saliency signal across retention ratios and is selected as the final scoring method.

7. Alignment Diagnostic

Sections 6 and the ablation collectively show that *which* CLIP-derived signal we use barely matters: GradCAM, patch–text cosine similarity, and value-feature scoring all hover around the random baseline. This suggests the bottleneck is the *signal* itself, not its specific instantiation. We test this directly by measuring how well PAPIT's GradCAM scores predict LLaVA's own attention to image tokens.

Setup. For each of N=100 GQA testdev-balanced samples [11], we compute (i) PAPIT's per-patch GradCAM scores $s^{\text{clip}} \in \mathbb{R}^{576}$ on LLaVA's vision tower, and (ii) the average attention $s^{\text{llava}} \in \mathbb{R}^{576}$ paid by the first 8 generated tokens to each of the 576 image-token positions, taken from the last LLM layer averaged over heads. We report per-image Spearman rank correlation $\rho(s^{\text{clip}}, s^{\text{llava}})$ and the Jaccard overlap between their respective top-10% patch sets. A query-agnostic uniform-random selector yields top-10% Jaccard $\approx 58/(2{\cdot}576 - 58) \approx 0.053$ in expectation.

Result. We repeat the measurement on 100 samples from each of GQA (testdev-balanced), TextVQA (validation), and VQA v2 (validation). Table 4 shows the result is consistent across all three datasets: the mean (and median) ρ is statistically indistinguishable from zero, no sample exceeds $\rho = 0.3$, and nearly half are negatively correlated. On every dataset, the top-10% Jaccard falls *below* the uniform-random reference of 0.053, with TextVQA showing the lowest agreement (0.039) despite being the only dataset where PAPIT slightly out-scores random in end-task accuracy (Table 2). This rules out the intuitive explanation that PAPIT's TextVQA gain stems from better CLIP–LLM alignment, and is consistent instead with the gain being within run-to-run noise of a near-random selector.

Table 4. Alignment between PAPIT GradCAM scores and LLaVA's last-layer LLM attention over the 576 image tokens (100 samples per dataset, LLaVA-1.5-7B). All three datasets give $\rho \approx 0$ and top-10% Jaccard below the uniform-random reference (0.053).

Metric	GQA	TextVQA	VQA v2
Spearman ρ mean	-0.018	-0.016	0.014
Spearman ρ median	0.008	0.019	0.010
Spearman ρ std	0.135	0.176	0.092
Frac. $\rho > 0.3$	0.0%	0.0%	0.0%
Frac. $\rho < 0$	48.0%	47.0%	43.0%
Top-10% Jaccard	0.047	0.039	0.046

These numbers are the empirical core of this paper. They imply that the CLIP-aligned saliency we feed into PAPIT is, at the patch level, indistinguishable from noise with respect to what the LLM actually attends to, on every benchmark we tested. Any training-free pruner that scores patches in a CLIP-aligned space — regardless of whether scoring uses cosine similarity, GradCAM, or value-feature norms — will be subject to the same ceiling and cannot, on principle, exceed the random baseline by more than the small sampling-noise band where CLIP saliency happens to coincide with LLM attention by chance.

Positive control: an LLM-internal-attention oracle. A natural follow-up is whether a *different* training-free signal — specifically LLaVA's own LLM attention — can beat random under the same pruning pipeline. We test a two-pass oracle: run LLaVA once on the full 576-token sequence, sum non-image-query attention to each image-token key at a given layer (averaged over heads), use that as the patch score, then re-generate on the spliced (pruned) inputs. This is strictly more expensive than single-pass FastV [7] but isolates the question of whether the *signal* can carry information that the pruning pipeline can exploit.

Table 5. Two-pass LLM-attention oracle on GQA (100 samples, exact match). **Bold** = beats random. L = 16 attains 0.61 at $k = 25\%$ (+0.08 vs. random, -0.07 vs. unpruned), and matches unpruned at $k = 50\%$. FastV's recommended layer 2 does not transfer to the two-pass setting.

Selector	$k = 25\%$	$k = 50\%$	$k = 75\%$
Unpruned	0.58	0.58	0.58
Random	0.53	0.53	0.56
PAPIT (CLIP)	0.56	0.53	0.51
LLM-attn $L = 2$	0.49	0.54	0.53
LLM-attn $L = 8$	**0.58**	**0.54**	0.56
LLM-attn $L = 16$	**0.61**	**0.58**	0.56
LLM-attn $L = 24$	**0.60**	**0.57**	0.55
LLM-attn $L = 31$	0.50	**0.55**	0.55

Table 5 shows that mid-layer LLM-attention (L = 8 to 24) consistently beats random, while early (L = 2) and late (L = 31) layers do not. Two readings follow. (i) The negative result of Table 4 is specifically about *CLIP-aligned* signals: a different training-free signal *can* carry useful patch importance information, so prompt-aware pruning is not impossible in principle. (ii) The signal lives in the *middle* of the LLM, not where FastV [7] recommends ($L = 2$ in their single-pass setting). In our two-pass setting, $L = 2$ has not yet seen enough of the prompt to commit attention mass to query-relevant patches, so its scores are essentially positional. We caveat that $N = 100$ gives a binomial standard error of ≈ 0.05, so the L = 16 advantage at $k = 25\%$ (+0.08 vs. random) is suggestive rather than conclusive; we leave larger-N replication and cross-dataset validation to future work.

8. Discussion

The alignment diagnostic of Section 7 explains *why* PAPIT cannot beat random pruning, but it does not by itself explain how PAPIT manages to nearly *match* random pruning despite using a near-noise scoring signal. We discuss this asymmetry, the one regime where prompt-awareness still helps, and what the negative result rules in or out for future work.

Why PAPIT does not collapse to far-below-random accuracy. LLaVA's LLM exhibits substantial redundancy across its 576 visual tokens: prior work [7] reports that more than half of the visual tokens can be dropped after layer 2 with negligible accuracy loss. Combined with LLaVA's strong language prior, this means the dominant accuracy driver at $k \geq 50\%$ is simply the *number* of retained tokens, not their identity. Our results are consistent with this: PAPIT, random, and (in the ablation) value-feature scoring all sit within

$\sim$1 point of each other at k=75% on GQA and VQA v2.

Why random is in fact a strong baseline. Two further effects work in random's favour. First, uniform sampling preserves *spatial coverage* of the 24×24 grid, whereas PAPIT concentrates retained tokens in a contiguous high-saliency region (Figure 2), discarding global cues that VQA questions often implicitly rely on. ViTs are also known to use some patch tokens for global register-like roles [14], which uniform sampling preserves in expectation. Second, top-k selection is redundancy-prone: nearby high-GradCAM patches encode correlated information, motivating the similarity-aware merging of ToMe [15] and the diversity-preserving hybrids of recent VLM-pruning work such as SparseVLM [16].

The TextVQA +0.9 is not a real win. We initially read the TextVQA k=75% gap (+0.9 over random) as evidence that prompt-awareness pays off when answer-bearing regions are small and spatially localized [13]. The alignment diagnostic refutes this reading: TextVQA in fact has the *lowest* CLIP–LLM Jaccard of the three datasets we tested (0.039, vs. 0.047 on GQA and 0.046 on VQA v2). With the underlying signal statistically equivalent across datasets, the ±1 point fluctuations between PAPIT and random in Table 2 are best read as run-to-run noise of two near-equivalent selectors operating on a redundant token set, not as a genuine prompt-aware advantage in any single regime.

Implications for future work. The diagnostic rules out CLIP-aligned saliency as a viable training-free signal. The positive control (Table 5) points the way out: mid-layer LLM attention does carry usable information, so the productive direction is to make that signal *cheap* to read. Two concrete options follow. First, distil mid-layer LLM attention into a small predictor over LLaVA's own post-instruction-tuning hidden states, in the spirit of the learned predictor in DynamicViT [5] but conditioned on the user prompt and supervised by attention rollout from layers $\sim$8–24 [17]. Second, integrate single-pass FastV-style pruning [7] with our ViT-side pipeline: apply a coarse CLIP-blind pre-prune (e.g., ToMe-style merging [15]) before the LLM, and a fine FastV-style cut inside the LLM, with the latter operating at the layer where the signal is informative rather than the canonical layer 2. Spatial-diversity priors (grid-stratified selection, DPP-style [18] relevance–diversity sampling) become meaningful only once a non-trivial relevance signal is in hand; on a near-noise score they cannot recover what the signal does not contain.

9. Conclusion

We set out to test whether a training-free, prompt-conditioned saliency signal derived from LLaVA's own CLIP-initialized vision tower could pick image tokens well enough to beat a query-agnostic random baseline. Across three VQA benchmarks (700 samples each), PAPIT matches but does not exceed random pruning, despite producing visibly localised selections on individual examples. An alignment diagnostic on 100 samples per dataset shows why: GradCAM saliency and LLaVA's internal LLM attention over the 576 image tokens are statistically uncorrelated on every benchmark (Spearman $\rho \approx 0$, top-10% Jaccard *below* the uniform-random reference). The CLIP-aligned signal that the entire family of training-free prompt-aware methods relies on therefore carries essentially no information about which patches the LLM actually uses — a constraint that no scoring rule within that signal family can overcome.

A two-pass LLM-internal-attention oracle, run through the same pruning pipeline, paints a complementary picture: *mid*-layer attention ($L \in [8, 24]$ for LLaVA-1.5-7B) does beat random and recovers up to 0.61 at $k = 25\%$ on GQA, while the early ($L = 2$) and late ($L = 31$) layers do not. Prompt-aware pruning is therefore achievable, but only when the relevance signal is read from *inside* the LLM, at the right depth.

Our contributions are accordingly threefold. (i) A clean training-free pipeline with 41–90% attention-FLOPs reduction and $\leq$2.3-point accuracy loss vs. unpruned on GQA and VQA v2 at k=75%. (ii) A cross-dataset negative result on training-free CLIP-aligned saliency. (iii) A positive control identifying mid-layer LLM attention as a usable training-free signal, motivating distillation into a small learned predictor over LLaVA hidden states or a hybrid ViT-side / FastV-style two-stage pruner that operates at the informative depth rather than the canonical layer 2.

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